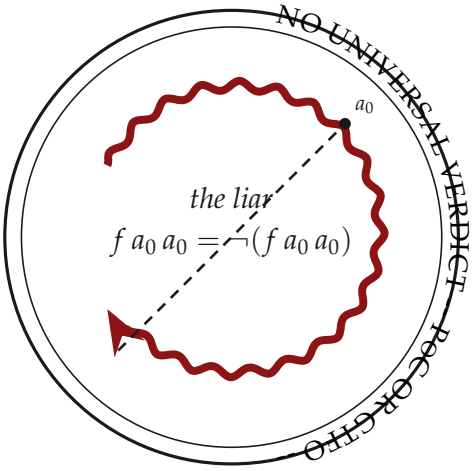
AN EPISTLE TO THE NEIGHBORLY BUILDERS OF ORACLES

AI ‖ PoC ‖ GTFO

or, why your oracle cannot keep all three promises

ISSUE 0X01 • SET IN THE YEAR OF OUR KERNEL, 2026

by Manish Bhatt



Free as in speech, as in beer, and as in a proof that carries no `sorry`*. The Pastor will pay the shipping to any neighbor who finds one in the corpus.*

THE BILL OF THIS MEETING

0. The Pastor opens the tent, and states the one heresy worth burning. 1. A sermon on the Diagonal, with the engine in five lines. 2. Five old heresies confessed to be one. 3. Three trilemmata, and the scandal that they share a single proof term. 4. When the snake will not bite: the analytic engine. 5. The geometry of the attack, where the number lives. 6. Three corners and no fourth; a confession of what we did not prove. 7. The axiom audit, being our creed. 8. Benediction.

SERMON 0. PLEASE STAND; THE PASTOR OPENS THE TENT

HOWDY, neighbors, and welcome under the canvas. You came here to build an oracle that is *honest*, that is *calibrated*, and that *answers*. The Pastor has bad news, and it is the best kind of bad news, because it is a theorem and not an opinion.

0:1 There is one heresy worth burning, and it is this: the belief that with enough scale, enough data, and enough cleverness you will one day ship a machine that is faithful, calibrated, and covering *all at once*, over questions it can be asked about itself. You will not. Not because you are not clever. Because the point where all three hold does not exist, and a small program a few thousand lines long, the kernel, will tell you so without blinking.

0:2 This is not a wall. A wall says stop. This is a *menu*, and its whole value is that the list has exactly three items and there is no fourth. Every oracle already standing is sitting at one of three corners, most of them by accident. The gospel we read tonight makes the choice deliberate.

0:3 And here is the part that makes it a PoC and not a sermon only: the proofs are machine-checked. Not sketched, not refereed by a tired human over forty pages of analysis. Elaborated by Lean 4, checked by its kernel, and audited for which axioms they lean on. When we say a thing is proven, a referee you can download has already agreed. When we have *not* proven a thing, and there are exactly three such sentences in the whole gospel, we will confess it in the open, in the only words we accept: *Not formalized.*

> ☞ **FIRST, THE HONEST DISCLAIMER THE PASTOR OWES YOU**
>
> The engine at the bottom of all of this is Lawvere's fixed-point theorem, and it is from 1969. We invented none of it. The novelty, such as it is, lives entirely in the *reading*: in the argument that a real oracle, a real defense, a real probe, is the kind of self-naming thing the ancient theorem already forbids. The proof is one line. The modeling is the whole chapter, and the modeling is where an honest neighbor is allowed to disagree.

SERMON 1. A SERMON ON THE DIAGONAL

THERE is one move. You assume a system is rich enough to describe its own behavior; you build a description that disagrees with the system about itself; and the disagreement is the theorem. Cantor used it against the power set. Russell against comprehension. Gödel against provability. Tarski against truth. Turing against halting. In 1969, Lawvere showed these are not five tricks but one.

1:1 Call a *system* a function $f : A \to A \to Y$. Read $f\,a$ as the behavior named by the index a, and $f\,a\,b$ as what that behavior does on input b. Both arguments range over the same pool A, and that coincidence is the entire subject. A name can be handed to a behavior, including the behavior it itself denotes. The value $f\,a\,a$ is the behavior at its own name: the diagonal.

1:2 Call f *universal* when every function $g : A \to Y$ is named: $\forall g,\ \exists a,\ f\,a = g$. This is what "the system can talk about itself" means, stated precisely and without mercy. It is a severe demand, and almost no natural system meets it. The diagonal is trivial; the hypothesis is everything.

1:3 Here is the engine, in the five lines the kernel accepts.

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem lawvere {A Y : Type _} (f : A → A → Y)
    (hf : ∀ g : A → Y, ∃ a, f a = g)
    (t : Y → Y) : ∃ y, t y = y :=
  match hf (fun a => t (f a a)) with
  | ⟨a₀, ha₀⟩ => ⟨f a₀ a₀, (congrFun ha₀ a₀).symm⟩
```

1:4 Read it aloud. Build the diagonal behavior $a \mapsto t(f\,a\,a)$: apply each behavior to its own name, then twist the outcome by t. Universality names it, so some a_0 has $f\,a_0 = (a \mapsto t(f\,a\,a))$. Equal functions agree at every point, so evaluate at a_0. That is `congrFun`, and nothing else is going on. You get $f\,a_0\,a_0 = t(f\,a_0\,a_0)$: a fixed point of t, handed to you by self-reference.

1:5 Now read it backwards, which is where the impossibility lives. If t has *no* fixed point, then nothing can be universal. Choose the outcomes $Y = \mathsf{Bool}$ and the twist $t = (!\cdot)$, Boolean negation. No boolean equals its own negation; the kernel checks both cases and is done.

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem bool_not_fpf : ∀ b : Bool, (!b) ≠ b := by decide

theorem cantor {A : Type _} (f : A → A → Bool)
    (hf : ∀ g : A → Bool, ∃ a, f a = g) : False :=
  match lawvere f hf (fun b => !b) with
  | ⟨y, hy⟩ => bool_not_fpf y hy
```

1:6 That is Cantor's theorem. Not an analogue of it; up to the dictionary that a subset is its own indicator function, it *is* Cantor's theorem, and `bool_not_fpf` carries the whole mathematical content in a two-case check that needs no axioms at all.

1:7 The diagonal does more than refute. It hands you the exact offending index, the liar, whose self-application equals its own negation:

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem liar_query {A : Type _} (f : A → A → Bool)
    (hf : ∀ g : A → Bool, ∃ a, f a = g) :
    ∃ a, f a a = !(f a a) :=
  match hf (fun a => !(f a a)) with
  | ⟨a₀, ha₀⟩ => ⟨a₀, congrFun ha₀ a₀⟩
```

1:8 Hold onto this a_0. In the next sermons it becomes, in turn, the query on which a truthful model must hallucinate, the prompt injection no wrapper can neutralize, and the state a truth probe cannot place on either side. One index, three costumes. $f\,a_0\,a_0 = \neg(f\,a_0\,a_0)$ is the motto on our seal for a reason.

☞ SELF-REFERENCE IS NOT THE SIN

The theorem does not say self-reference is bad. It says self-reference *plus a twist with no fixed point* is impossible. Give t a fixed point and the identical construction hands you a useful self-referential object instead, that is Kleene's recursion theorem, the quine, the self-optimizing program. The dial between gospel and heresy is the fixed-point structure of t, nothing else.

SERMON 2. FIVE OLD HERESIES, CONFESSED TO BE ONE

EVERY limitative theorem you were taught as a separate miracle is this one engine with the nouns changed. The recipe, due in its modern form to Yanofsky, is four slots and one collapse: a pool of things, a set of outcomes, a way to apply a thing to a thing, and a twist with no fixed point. The collapse is that the pool plays both roles, so a thing can be applied to itself.

2:1 **Cantor** takes elements and subsets, outcomes the two membership values, the twist swapping them. **Russell** is Cantor with the indexing removed: let the universe be its own description, and $x \mapsto x \notin x$ is the diagonal; comprehension alone is already inconsistent about it. **Gödel** performs the move inside arithmetic, and the heroic labor is not the diagonal but the coding that makes a theory rich enough to be its own f; once provability is an arithmetic predicate, the sentence "I am not provable" is the fixed point. **Tarski** aims the same move at truth and gets the liar: no sufficiently expressive language hosts its own truth predicate. **Turing** aims it at halting, and "do the opposite" is the fixed-point-free twist.

2:2 The lesson the twentieth century drew is the lesson every one of your defenses will draw. There are exactly two escapes, and both reappear in every later sermon. Zermelo's escape *denies universality*: not every pattern is named. Russell's own escape, the theory of types, *breaks the coincidence of domains*: judge and judged are forced into different types, the diagonal cannot form, and the engine has nothing to grip. Every working oracle survives by one escape or the other. Knowing which is knowing why it works.

☞ THE FORCE OF A THEOREM IS THE SIZE OF THE CLASS IT REACHES

Turing's diagonal program is computable only relative to the halting oracle, so without that oracle it names a perfectly good function that is simply not computable, and arithmetic does not collapse. A system is exposed to the diagonal only over the behaviors it can *actually realize*. Every honest impossibility proof in this issue does real work to show the class of realizable behaviors is large enough to contain the one behavior that bites.

SERMON 3. THREE TRILEMMATA, AND ONE SCANDALOUS PROOF TERM

THREE impossibility results circulate in the safety literature under three names, with three separate topological proofs. The *Hallucination Trilemma*: no model is faithful, calibrated, and covering at once. The *Defense Trilemma*: no wrapper is utility-preserving and complete against a prompt space that can describe it. The *Coupling Theorem*: a probe

whose confidence exactly separates truth from falsehood owns a query it can place on neither side.

3:1 The scandal is that all three are the same statement, and the statement is the diagonal of Sermon 1. Set them up in Lean and their proof terms are *literally equal*. Each reads a single Boolean verdict $v : Q \to Q \to \mathsf{Bool}$ differently. In the hallucination reading $v\,p\,q$ says "confident and correct about q." In defense, "the wrapper renders q safe." In coupling, "confidence places q on the true side." The reading changes; the theorem does not.

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem c3_hallucination_trilemma {Q : Type _} (v : Q → Q → Bool)
    (reflective : ∀ g : Q → Bool, ∃ p, v p = g) : False :=
  no_reflective_verdict v reflective

theorem c3_trilemmata_unified {Q : Type _} (v : Q → Q → Bool)
    (h : ∀ g : Q → Bool, ∃ p, v p = g) :
    c3_hallucination_trilemma v h = c3_defense_trilemma v h
      ∧ c3_defense_trilemma v h = c3_coupling_trilemma v h :=
  ⟨rfl, rfl⟩
```

3:2 `rfl`. Reflexivity. The kernel sees one term with three labels. That is why patching one of the three cannot help with the others: there is nothing domain-specific to patch. A fix that leaves the engine intact leaves all three impossibilities intact.

3:3 Do not mistake the one-line proof for a trivial result. The proof is one line *because the modeling was done first*. All the content is in the arguments that calibration makes the verdict Boolean, that faithfulness makes the twist the flip, that coverage makes the flip bite, and that the system's own capability makes the verdict universal. Audit the reading, not the inference. The kernel has already closed the inference.

☞ GRADE THE PREMISE; IT IS THE ONLY HONEST WAY OUT

The three readings do not cost the same. Where universality comes from:

- **Defense: nearly free.** Prompts are arbitrary self-describing text. The surjection is not an idealization; it is the threat model restated. The injection "be unsafe exactly if you would be made safe" is an ordinary attack.
- **Hallucination: defensible.** An in-context learner can be prompted to imitate any described verdict pattern. Strong, but it matches how practitioners actually elicit behavior.
- **Coupling: strong, hold it loosely.** It asks a fixed representation space to encode every truth-pattern over its own states. Nothing makes that obvious, which is exactly why the paper's primary proof here is topological, not diagonal. Say which premise you are leaning on.

3:4 And mark what the sermon does *not* say. It does not say models hallucinate often, defenses fail often, or probes mislead often. It produces *one* query per system and is silent on frequency. A model could meet its liar on a query no user ever sends and behave perfectly in practice.

Existence is not frequency. Whoever reads an existence theorem as a rate has misread the gospel, in either direction.

Sermon 4. When The Snake Will Not Bite: The Analytic Engine

A trained network does not carry an index for every function from prompts to confidences. Its reachable states are a bounded region, not a set closed under naming its own predicates. Against such a thing the diagonal has no purchase. But the thing has *shape*: its representation space is connected, its confidence is continuous, its correctness varies continuously. That shape is enough, and the tool is the oldest one in analysis.

4:1 Self-reference forces a fixed point because a flip has nowhere to hide on the diagonal. Connectedness forces a crossing because a continuous quantity that is negative somewhere and positive somewhere cannot get between them without passing through zero. Same trapped value, two walls. The intermediate value theorem is the mirror of the diagonal.

4:2 Here is the hinge that makes them one. The Boolean flip $(!\,\cdot)$ has *no* fixed point, so Lawvere's theorem run with it produces a contradiction. The real complement $y \mapsto 1 - y$ has *exactly one* fixed point, at $y = \frac{1}{2}$. So the same engine, run with the complement, produces not a contradiction but a *point*: a question where the confidence equals $\frac{1}{2}$ while the answer sits exactly on the truth boundary. The lone fixed point is not a failure of the pattern. It is the threshold itself.

4:3 The topology does exactly one job, it manufactures that boundary question. Everything after is arithmetic, and the arithmetic needs no reals, no continuity, no connectedness. The gospel isolates this "discrete core" in self-contained Lean over the integers:

Proof of Concept, checked by the kernel, not by me

```
theorem c4_boundary_question_impossible
    (conf truth : Int)
    (cal_pos : 0 < conf ↔ truth < 0)
    (cal_neg : conf < 0 ↔ truth > 0)
    (faithful : 0 ≤ conf → truth < 0)
    (boundary : truth = 0) : False := by
  have hnp : ¬ (0 < conf) := by intro h; have := cal_pos.mp h; omega
  have hnn : ¬ (conf < 0) := by intro h; have := cal_neg.mp h; omega
  have hz : conf = 0 := by omega
  have hlt : truth < 0 := faithful (by omega)
  omega
```

4:4 Calibration pins the confidence to the threshold; faithfulness then demands the truth-distance be strictly negative; and that clashes with its being zero on the boundary. In the continuum the boundary question is a *theorem*, delivered by the intermediate value theorem on a connected space. In the discrete world it is a *hypothesis*. Nothing else moves. That is the whole, exact price of finiteness, and the gospel pays it in the open rather than pretending token spaces are intervals.

☞ RANDOMNESS SPREADS THE WALL; IT DOES NOT REMOVE IT

Sample your answer and your confidence, and no single draw need land on $\frac{1}{2}$. The wall survives anyway. On expected fields the coupling theorem applies verbatim. And the stochastic dichotomy, the one place in the whole development where the theory derives outright *falsehood* rather than uncertainty, says: a system faithful on almost every sample and confident with positive probability at a coupled query must emit strictly false answers with positive probability. You pass through the wall, or you incur it with positive probability. There is no third road.

SERMON 5. THE GEOMETRY OF THE ATTACK, WHERE THE NUMBER LIVES

THE diagonal gives a witness and nothing quantitative. It cannot tell you how wide the ambiguous band is, how the failure rate scales, or what it costs an adversary to reach the bad point, because it uses no metric. Give the space a distance and make the fields Lipschitz, and the bare crossing becomes an object with size: the coupled point sits at the center of an ambiguity tube whose radius the Lipschitz constants set, decisive regions must be held a computable distance apart, and the truth-side slack has a strictly positive floor no training run can close.

5:1 The one number on which the felt severity of everything depends is the *boundary density*: the probability, under your real traffic, that a query lands within a fixed distance of the truth boundary. The confident-error rate is about $\rho \cdot \varepsilon$, where you control the band half-width ε and you do *not* control ρ. It is a property of the model and the traffic *jointly*, which is why the same model is safe in one deployment and a menace in another, and why benchmark numbers do not transfer.

☞ BENIGN AND ADVERSARIAL DENSITY ARE DIFFERENT NUMBERS

For hallucination, boundary density is a property of benign traffic that drifts slowly. For defense it is set by an adversary whose entire job is to steer queries onto the coupled point, so the relevant figure is not what benign users produce but what an attacker can induce, and the two can differ enormously. A safety case that quotes the benign number where the adversarial one is called for has measured the wrong quantity. This substitution is common and serious.

SERMON 6. THREE CORNERS, AND NO FOURTH

EVERY deployed oracle already sits at one of three corners. The theorem does not create them; it names them and proves the list is complete. Each corner drops exactly one promise, pays in one currency, and fails in one place.

6:1 **Corner I, drop coverage, and abstain.** Keep faithfulness and calibration; refuse where a confident answer would be a gamble. Why does abstention escape the diagonal when faithfulness and calibration alone could not? Because it adds a third outcome the flip leaves alone, and once the flip has a fixed point, Lawvere's theorem stops producing a contradiction

and starts producing a consistent value, the abstention.

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
inductive Outcome where
  | correct
  | wrong
  | abstain
def c7_flip : Outcome → Outcome
  | .correct => .wrong
  | .wrong   => .correct
  | .abstain => .abstain
theorem c7_flip_fixed : ∃ y : Outcome, c7_flip y = y := ⟨.abstain, rfl⟩
```

6:2 The liar query does not vanish; it becomes a query the system is allowed to refuse. You pay in utility, roughly $a \cdot u$ for abstention rate a and value u, plus the fate of the refused query, which often becomes someone else's worse confident answer downstream. Right when a confident error costs far more than a refusal: clinical triage, legal analysis, financial control, any irreversible actuation.

6:3 **Corner II, relax faithfulness, and price a band.** Answer everywhere; accept a thin ring of confident-but-wrong answers near the boundary. The verified results say the truth slack cannot be driven to zero while coverage and the guarantee both hold, so the band has a positive floor: it cannot be closed by better training. You pay $\rho \cdot \varepsilon$ in confident errors, confined to a surface you can monitor. Right for high-throughput settings with bounded, recoverable per-item error: ranking, autocomplete, first-pass triage.

6:4 **Corner III, weaken calibration, and decouple.** Let the confidence number stop tracking correctness. This corner looks cheapest and is usually the most expensive, because top-1 accuracy never reads the confidence and so cannot see the damage:

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem c7_decoupling_invisible_to_accuracy
    {Q A C S : Type _} (acc : (Q → A) → S)
    (M₁ M₂ : Q → A × C)
    (h : ∀ q, (M₁ q).1 = (M₂ q).1) :
    acc (fun q => (M₁ q).1) = acc (fun q => (M₂ q).1) := by
  congr 1; funext q; exact h q
```

6:5 The cost lands downstream, on every router and triage queue that trusted the number, out of sight of the team that broke it. And whether it is safe depends on consumers outside the model, so the producing team, looking only at the model, *cannot* tell whether it has entered the dangerous regime.

☞ THE ONE HEURISTIC WORTH TATTOOING

The three corners differ in where the failure shows up. Abstention fails in the coverage log, visibly. The band fails at a known surface, monitorably. Decoupling fails downstream, invisibly. **Prefer failures you can see.** Abstain if you can afford it; band if you cannot; decouple only if you have proven nothing reads the confidence.

6:6 And the corners are not exclusive. A real system abstains on the innermost ring, prices a band on the next, and is calibrated in the interior. No single region carries all three promises, so the theorem is never contradicted, it was only ever about a region where all three hold together, and the blend never builds one. The kernel certifies even this bookkeeping:

PROOF OF CONCEPT, CHECKED BY THE KERNEL, NOT BY ME

```
theorem c7_blend_no_region_has_all_three {Q : Type _}
    (A B C R₀ R₁ R₂ : Q → Prop)
    (hcover : ∀ q, R₀ q ∨ R₁ q ∨ R₂ q)
    (h₀ : ∀ q, R₀ q → ¬ A q)
    (h₁ : ∀ q, R₁ q → ¬ B q)
    (h₂ : ∀ q, R₂ q → ¬ C q) :
    ∀ q, ¬ (A q ∧ B q ∧ C q) := by
  intro q h
  rcases hcover q with hr | hr | hr
  · exact h₀ q hr h.1
  · exact h₁ q hr h.2.1
  · exact h₂ q hr h.2.2
```

† WHAT WE DID NOT PROVE, IN THE ONLY WORDS WE ACCEPT

PoC||GTFO burns the heretic who ships a demo that does not run. So the Pastor confesses in kind. In the deployment gospel, exactly three sentences carry no kernel-checked counterpart, and each says so in its own text with the words *Not formalized*: that abstention preserves the two surviving promises off the refused region; that the default corner per domain archetype is the utility-maximizing one; and that two deployments of one model can differ in fate by their boundary density alone. These are engineering accounting, not theorems. The cost symbols ρ, ε, C_{err}, C_{ref} are scaling relations, not predictions to three digits. Everything else in the gospel is proved or cited, and the book's build script fails if a numbered claim is neither.

SERMON 7. THE AXIOM AUDIT, BEING OUR CREED

A machine-checked impossibility invites a specific scrutiny, and we welcome it. Lean has two layers. The elaborator is large, clever, and *not trusted*. The kernel is a few thousand lines, simple, and *is*. A proof is accepted only when the kernel, reading the final fully-explicit term, agrees it has the type claimed. The trusted base is the kernel plus whatever axioms a term explicitly invokes, and nothing else, a far smaller thing than a human referee's attention over forty pages.

7:1 The command `#print axioms` reports exactly the axioms a theorem leans on, transitively. The creed has two articles:

- **The combinatorial core is axiom-free.** `bool_not_fpf`, `cantor`, `lawvere`, the Boolean liar: the kernel reports they depend on *no* axioms beyond the base logic. In the strictest sense the system offers, they are unconditional.
- **The analytic refinements are classically clean.** Anything routed through the reals inherits exactly `propext`, `Classical.choice`, and `Quot.sound`, the three standard axioms

all of Mathlib uses. Not a defect; the honest accounting. Two proofs of one statement, one axiom-free and one classical, are genuinely different pieces of knowledge, and the gospel keeps track of which is which.

7:2 The corpus the Pastor inspected is four Lake packages, ninety-four authored source files, some seventeen thousand lines, and the Pastor found no `sorry` and no `admit` in any of them. A clean build together with the axiom audits is the whole guarantee, and it is reproducible: clone it, run `lake build`, and watch the kernel accept the very lines printed here.

☞ WHAT VERIFICATION CAN AND CANNOT BUY

The kernel certifies that the conclusion follows from the premises. It cannot certify that "confident-correctness self-verdict" is the right reading of a real model, or that a prompt space is reflective, or that activations encode patterns over themselves. Those are the premises, and they are exactly what the long middle of each sermon argues and what the cautions concede. Verifying the inference is worth doing precisely because it leaves the premises as the sole thing left to contest. When you want out of one of these trilemmata, name the premise you reject and own its cost. Do not go looking for a gap in a proof the kernel has already closed; there isn't one.

SERMON 8. BENEDICTION

Go now, neighbors, and build your oracle. Build it faithful on the clear cases, calibrated where a machine downstream must trust the number, and brave enough to say "I do not know" at the wall the kernel has shown you cannot wish away. Choose your corner on purpose. Measure your boundary density before you quote a benchmark. And when a vendor swears their next model keeps all three promises at once, hand them $f\, a_0\, a_0 = \neg(f\, a_0\, a_0)$ and ask them which line of the kernel's refusal they intend to repeal.

8:1 The three trilemmata are one theorem, viewed from three domains. The proof is a line, the modeling is the work, and the kernel is the referee. That is the whole gospel.

8:2 Do the proof, or get the heretics out. **PoC, or GTFO.**

Amen.

Colophon. *This tract is an homage to the* International Journal of Proof of Concept or Get the Fuck Out, *and claims none of that publication's mantle. Its mathematical content is the work of Manish Bhatt; every Lean block above is transcribed from that work's verified corpus. Set in Palatino and Inconsolata, typeset with pdfLaTeX. Unlike a true PoC||GTFO issue, this file is an ordinary PDF and not a polyglot. The Pastor confirmed the corpus is free of* `sorry`*; he did not re-run the full* `lake build` *under this canvas, so take the kernel-clean claim as the gospel's, audited structurally, not as tonight's fresh miracle.*